Probability and Statistics

Random Sampling Point Estimation

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A population is considered to be known when

we know the probability distribution $f(x)$ of the associated random variable X.

If, for example, X is normally distributed, we say that the population is normally distributed or that we have a normal population. Similarly, if X is binomially distributed, we say that the population is binomially distributed or that we have a binomial population.

Population Parameters

There will be certain quantities that appear in $f(x)$, such as μ and σ in the case of the normal distribution or p in the case of the binomial distribution. Other quantities such as the median, moments, and skewness can then be determined in terms of these. All such quantities are often called population parameters.

When we are given the population so that we know $f(x)$, then the population parameters are also known.

An important problem arises when the probability distribution $f(x)$ of the population is not known precisely.

Although we may have some idea of, or at least be able to make some hypothesis concerning, the general behavior of $f(x)$.

For example, we may have some reason to suppose that a particular population is normally distributed.

In that case we may not know one or both of the values μ and σ and so we might wish to draw statistical inferences about them.

In the field of **statistical inference**, statisticians are interested in arriving at conclusions concerning a population when it is impossible or impractical to observe the entire set of observations that make up the population.

For example, in attempting to determine the average length of life of a certain brand of light bulb, it would be impossible to test all such bulbs if we are to have any left to sell. Exorbitant costs can also be a prohibitive factor in studying an entire population. Therefore, we must depend on a subset of observations from the population to help us make inferences concerning that same population.

This brings us to consider the notion of **sampling**.

Draw conclusions about the heights of 12,000 adult students (the population) by examining only 100 students (a sample) selected from this population.

To obtain a sample of size 100,

1. First choose one individual at random from the population, say, x_1 the value of a random variable X_1
2. Choose the second individual for the sample, who can have any one of the values x_2 of the possible heights, and x_2 can be taken as the value of a random variable X_2.

We can continue this process up to X_{100} since the sample size is 100.

Alternatively

we can consider all possible samples of size n that can be drawn from the population, and for each sample we compute the statistic. In this manner we obtain the distribution of the statistic, which is its sampling distribution.

Sampling Distribution of Means

Let $f(x)$ be the probability distribution of some given population from which we draw a sample of size n.

Then it is natural to look for the probability distribution of the sample statistic $\overline{X}$ which is called the **sampling distribution** for the sample mean, or the sampling distribution of means.

The **mean** of the sampling distribution of means, denoted by $\mu_{\overline{X}}$ is given by

$$E[\overline{X}] = \mu_{\overline{X}} = \mu$$

where μ is the mean of the population.

Suppose that the population consist of five numbers X=1,2,3,4,5. Draw all possible sample with replacement of size 2 and also verify that $\mu_{\overline{X}} = \mu$.

As given that $N = 5$ and $n = 2$ it means that sample size is $N^n = 25$.
$(1,1),(1,2),(1,3),(1,4),(1,5),(2,1),(2,2),\cdots,(5,4),(5,5)$
The corresponding means with probabilities are

$\overline{X}$	1	1.5	2	2.5	3	3.5	4	4.5	5
f($\overline{X}$)	1/25	2/25	3/25	4/25	5/25	4/25	3/25	2/25	1/25

$$\mu = \frac{\sum X}{N} = \frac{1+2+3+4+5}{5} = 3$$

$$\mu_{\overline{X}} = \sum \overline{X} f\left(\overline{X}\right) = \frac{1 + (1.5)(2) + \cdots + 5}{25} = 3$$

If a population is infinite and the sampling is random or if the population is finite and sampling is with replacement, then the variance of the sampling distribution of means, denoted by $\sigma^2_{\overline{X}}$ is given by

$$E[(\overline{X} - \mu)^2] = \sigma^2_{\overline{X}} = \frac{\sigma^2}{n}$$

where σ^2 is the variance of the population.

Standard error of the mean

When the sample size is less than 5% of the population size, the standard error in the mean is

$$\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}$$

$$\overline{X} = \sum_{i=1}^{n} \frac{X_i}{n} \quad Var(\overline{X}) = Var\left(\sum_{i=1}^{n} \frac{X_i}{n}\right)$$

$$Var(\overline{X}) = \frac{Var\left(\sum_{i=1}^{n} X_i\right)}{n^2} = \frac{1}{n^2}\left[Var\left(X_1 + X_2 + \cdots + X_n\right)\right]$$

$$Var(\overline{X}) = \frac{1}{n^2}\left[\sigma^2 + \sigma^2 + \cdots + \sigma^2\right]$$

$$Var(\overline{X}) = \frac{1}{n^2}(n\sigma^2) = \frac{1}{n}\sigma$$

$$\boxed{\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}}$$

Let mean and variance of the population of μ and σ

The mean and variance of the sampling distribution of means, denoted by $\mu_{\overline{X}}$ and $\sigma^2_{\overline{X}}$ are given by

$$\mu_{\overline{X}} = \mu, \quad \text{and} \quad \sigma^2_{\overline{X}} = \frac{\sigma^2}{n}$$

The standardized variable associated with $\overline{X}$ is given by

$$Z = \frac{\overline{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

A point estimate of some population parameter θ is a single value $\hat{\theta}$ of a statistic $\hat{\Theta}$.

For example, the value $\bar{x}$ of the statistic $\bar{X}$, computed from a sample of size n, is a point estimate of the population parameter μ.

Three Properties of a Good Estimator

1. The estimator should be an **unbiased** estimator. That is, the expected value or the mean of the estimates obtained from samples of a given size is equal to the parameter being estimated.
2. The estimator should be **consistent**. For a consistent estimator, as sample size increases, the value of the estimator approaches the value of the parameter estimated.
3. The estimator should be a **relatively efficient** estimator. That is, of all the statistics that can be used to estimate a parameter, the relatively efficient estimator has the smallest variance.

Unbiasedness

An estimator is defined to be ***unbiased*** if the statistic used as an estimator has its expected value equal to the true value of the of the population parameter being estimated.

Let $\hat{\theta}$ be the estimator of the parameter θ then $\hat{\theta}$ will be called unbiased estimator if

$$E[\hat{\theta}] = \theta$$

1. If $E[\hat{\theta}] \neq \theta$ then the statistic is said to be a ***biased estimator.***
2. The estimator is said to be positively biased when $E[\hat{\theta}] - \theta > 0$.
3. It is said negatively biased if $E[\hat{\theta}] - \theta < 0$.

If $\hat{\theta} = \overline{X}$ and $\theta = \mu$ then show that it is unbiased estimator.

Examples

$\overline{X}$ is unbiased estimator of a

1. population mean μ i.e. $E[X] = \mu$
2. Bernoulli distribution parameter p i.e. $E[X] = p$
3. normal distribution parameter μ
4. a Poisson distribution λ

Let $X_1, X_2, \cdots, X_n$ be a random sample from the population with a mean of μ and a variance of σ^2. Then show that the sample variance $S^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$ is a biased estimator of σ^2.

$$\sum_{i=1}^{n}(X_i-\mu)^2 = \sum_{i=1}^{n}[(X_i-\bar{X})+(\bar{X}-\mu)]^2$$

$$= \sum_{i=1}^{n}(X_i-\bar{X})^2 + \sum_{i=1}^{n}(\bar{X}-\mu)^2 + 2(\bar{X}-\mu)\sum_{i=1}^{n}(X_i-\bar{X})$$

$$= \sum_{i=1}^{n}(X_i-\bar{X})^2 - n(\bar{X}-\mu)^2.$$

$$E(S^2) = E\left[\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2\right] = \frac{1}{n-1}\left[\sum_{i=1}^{n}E(X_i-\mu)^2 - nE(\bar{X}-\mu)^2\right]$$

$$= \frac{1}{n-1}\left(\sum_{i=1}^{n}\sigma_{X_i}^2 - n\sigma_{\bar{X}}^2\right) \boxed{\sigma_{X_i}^2 = \sigma^2, \text{ for } i = 1, 2, \ldots, n, \text{ and } \sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}}$$

$$E[S^2] = \frac{1}{n-1}\left(n\sigma^2 - n\frac{\sigma^2}{n}\right) = \sigma^2$$

Although S^2 is an unbiased estimator of σ^2, S, on the other hand, is usually a biased estimator of σ, with the bias becoming insignificant for large samples. This example illustrates why we divide by $n1$ rather than n when the variance is estimated.

Variance of a Point Estimator

If $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are two unbiased estimators of the same population parameter θ, we want to choose the estimator whose sampling distribution has the smaller variance.
Hence, if

$$\sigma_{\hat{\Theta}_1} < \sigma_{\hat{\Theta}_2}$$

we say that $\hat{\Theta}_1$ is a more efficient estimator of θ than $\hat{\Theta}_2$.

The Likelihood Function

There are many situations in which it is not at all obvious what the proper estimator should be. Here we deal with the method of maximum likelihood.

- The method of maximum likelihood is that for which the likelihood function is maximized.

Denote by $X_1, X_2, ..., X_n$ the independent random variables taken from a discrete probability distribution represented by $f(x, \theta)$ where θ is a single parameter of distribution. Now

$$L(x_1, x_2, \cdots, x_n; \theta) = f(x_1, x_2, \cdots, x_n; \theta) = f(x_1, \theta)f(x_2, \theta) \cdots f(x_n, \theta)$$

is the *joint distribution of the random variables*, often referred to as the ***likelihood function***.

$$L(\theta) = \prod_{i=1}^{n} f(x_i, \theta)$$ likelihood function is θ, not x.

In the case of a discrete random variable, the interpretation is very clear. The quantity $L(x_1, x_2, ..., x_n; \theta)$, the likelihood of the sample, is the following joint probability:

$$P(X_1 = x_1, X_2 = x_2, ..., X_n = x_n|\theta),$$

which is the probability of obtaining the sample values $x_1, x_2, ..., x_n$. For the discrete case, the maximum likelihood estimator is one **that results in a maximum value** for this joint probability or maximizes the likelihood of the sample.

Example

Lets consider a sample of size 10 from the binomial population with unknown parameter p. Suppose that $X = 4$, estimate the unknown parameter p

By the binomial formula, the probability of $X = 4$ when $n = 10$, given p, is

$$f(4, p) = C_4^{10} p^4 (1 - p)^6$$

Evaluating the expression for all possible values of p, we obtain

p	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
f(p)	0	.011	.088	.200	.251	.205	.111	.036	.005	0	0

Of all the values of p, the probability of the actual sample data corresponding to the value $p = 0.4$ is the greatest. In other words $p = 0.4$ makes the sample result most likely.

Now we wish to estimate the unknown parameter θ by that value which maximizes the the *likelihood function*$L(\theta)$. Such a value, called the ***ML estimator*** is obtained as follows

$$\frac{\partial L(\theta)}{\partial \theta} = 0$$

The value of θ obtained from above equation must satisfy

$$\frac{\partial^2 L(\theta)}{\partial \theta^2} < 0$$

For convenience, we find maximum of $\ln L(\theta)$, i.e.

$$\frac{\partial \left(\ln L\left(\theta\right)\right)}{\partial \theta} = 0 \qquad \frac{\partial^2 \left(\ln L\left(\theta\right)\right)}{\partial \theta^2} < 0$$

Consider the geometric distribution

$$f(x) = p(1-p)^{x-1} \qquad \textit{for } x = 1, 2, 3, \ldots$$

Find the maximum likelihood estimator for p when

(i) only one experiment is performed,

(ii) n experiments are performed.

(i)Since only one experiment is performed so

$$L(p) = f(x;p) = p(1-p)^{x-1}, \implies \ln L(p) = \ln p + (x-1)\ln(1-p)$$

Differentiating w.r.t p and equating it to zero, we obtain

$$\frac{\partial(\ln L(p))}{\partial p} = \frac{1}{p} - \frac{x-1}{1-p} = 0, \text{ or } 1 - p - px + p = 0.$$

It gives $\hat{p} = 1/x$. The likelihood function attains its maximum value at $p = 1/x$. Therefore, the maximum likelihood estimator (MLE) for parameter p is $p = 1/X$.

(ii) The likelihood function for n experiments would be

$$L(p) = \prod_{i=1}^{n} p(1-p)^{x_i - 1} = p^n (1-p)^{\sum_{i=1}^{n}(x_i - 1)} = p^n (1-p)^{\sum_{i=1}^{n} x_i - n}$$

$$\ln L(p) = n \ln p + \left(\sum_{i=1}^{n} x_i - n \right) \ln(1-p)$$

Differentiating w.r.t p and equating it to zero, we obtain

$$\frac{(1-p)\, n - p \left(\sum_{i=1}^{n} x_i - n \right)}{p(1-p)} = 0$$

$$n - p \sum_{i=1}^{n} x_i - pn + np = 0, \implies p = \sum_{i=1}^{n} \frac{n}{x_i}$$

$\hat{p} = 1/\bar{x}$.

Let $X_1, X_2, \cdots, X_n$ be a random sample from the Poisson distribution

$$f(x, \lambda) = \frac{\lambda^x}{x!} e^{-\lambda}, \qquad \text{Find the MLE of } \lambda.$$

$$L(\lambda) = \prod_{i=1}^{n} \frac{\lambda^{x_i}}{x_i!} e^{-\lambda}, \implies \ln L(\lambda) = \ln\left(\lambda^{\sum_{i=1}^{n} x_i} e^{-n\lambda}\right) - \ln\left(\prod_{i=1}^{n} x_i!\right)$$

$$\ln L(\lambda) = \ln\left(\lambda^{\sum_{i=1}^{n} x_i}\right) + \ln\left(e^{-n\lambda}\right) - \ln\left(\prod_{i=1}^{n} x_i!\right)$$

Differentiating w.r.t p and equating it to zero, we obtain

$$-n + \frac{\sum_{i=1}^{n} x_i}{\lambda} = 0, \qquad \lambda = \frac{\sum_{i=1}^{n} x_i}{n} = \overline{x}.$$

Find maximum likelihood estimates for $\theta_1 = \mu$ and $\theta_2 = \sigma$ in the case of the normal distribution.

Even the most efficient unbiased estimator is unlikely to estimate the population parameter exactly. It is true that estimation accuracy increases with large samples, but there is still no reason we should expect a point estimate from a given sample to be exactly equal to the population parameter it is supposed to estimate.

There are many situations in which it is preferable to determine an interval within which we would expect to find the value of the parameter. Such an interval is called an **interval estimate**.

An estimate of a population parameter given by a **single** number is called a **point estimate** of the parameter.

An estimate of a population parameter given by **two numbers** between which the parameter may be considered to lie is called an **interval estimate** of the parameter.

Example

If we say that a distance is 5.28 feet, we are giving a point estimate. If, on the other hand, we say that the distance is 5.28 ± 0.03 feet, i.e., the distance lies between 5.25 and 5.31 feet, we are giving an interval estimate.

The **confidence level** of an interval estimate of a parameter is the **probability** that the interval estimate will contain the parameter, assuming that a large number of samples are selected and that the estimation process on the same parameter is repeated.

A **confidence interval** is a **specific interval** estimate of a parameter determined by using data obtained from a sample and by using the specific confidence level of the estimate.

Let $\mu_{\overline{X}}$ and $\sigma_{\overline{X}}$ be the mean and standard deviation of the sampling distribution of a statistic $\overline{X}$. Then, if the sampling distribution of $\overline{X}$ is approximately normal, we can expect to find $\overline{X}$ lying in the intervals $\mu_{\overline{X}} - \sigma_{\overline{X}}$ to $\mu_{\overline{X}} + \sigma_{\overline{X}}$, $\mu_{\overline{X}} - 2\sigma_{\overline{X}}$ to $\mu_{\overline{X}} + 2\sigma_{\overline{X}}$ or $\mu_{\overline{X}} - 3\sigma_{\overline{X}}$ to $\mu_{\overline{X}} + 3\sigma_{\overline{X}}$ about 68.27%, 95.45%, and 99.73% of the time, respectively.

Equivalently we can expect to find, or we can be **confident** of finding $\mu_{\overline{X}}$, in the intervals $\overline{X} - \sigma_{\overline{X}}$ to $\overline{X} + \sigma_{\overline{X}}$, $\overline{X} - 2\sigma_{\overline{X}}$ to $\overline{X} + 2\sigma_{\overline{X}}$ or $\overline{X} - 3\sigma_{\overline{X}}$ to $\overline{X} + 3\sigma_{\overline{X}}$ about 68.27%, 95.45%, and 99.73% of the time, respectively.

The end numbers of these intervals ($\overline{X} \pm \sigma_{\overline{X}}$, $\overline{X} \pm 2\sigma_{\overline{X}}$, $\overline{X} \pm 3\sigma_{\overline{X}}$) are then called 68.27%, 95.45%, and 99.73% **confidence limits**.

Similarly, $\overline{X} \pm 1.96\sigma_{\overline{X}}$ and $\overline{X} \pm 2.58\sigma_{\overline{X}}$ and are 95% and 99% (or 0.95 and 0.99) **confidence limits** for $\mu_{\overline{X}}$. The percentage confidence is often called the **confidence level**. The numbers 1.96, 2.58, etc., in the confidence limits are called **critical values**, and are denoted by z_c .

Confidence Limits	Confidence Level	Critical Values z_c	Significance level α
$[\overline{X} - \sigma_{\overline{X}}, \overline{X} + \sigma_{\overline{X}}]$	68%		
$[\overline{X} - 2\sigma_{\overline{X}}, \overline{X} + 2\sigma_{\overline{X}}]$	95%	1.96	0.05
$[\overline{X} - 3\sigma_{\overline{X}}, \overline{X} + 3\sigma_{\overline{X}}]$	99%	2.58	0.01

Formula for the Confidence Interval of the Mean for a Specific α

$$\overline{X} - Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \overline{X} + Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

The term $Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$ is called the **maximum error of the estimate.**

α	C.L.	z_c	Conf. Limits
0.1	90%	± 1.645	$[\overline{X} - 1.645\sigma_{\overline{X}}, \overline{X} + 1.645\sigma_{\overline{X}}]$
0.05	95%	± 1.96	$[\overline{X} - 1.96\sigma_{\overline{X}}, \overline{X} + 1.96\sigma_{\overline{X}}]$
0.01	99%	± 2.58	$[\overline{X} - 2.58\sigma_{\overline{X}}, \overline{X} + 2.58\sigma_{\overline{X}}]$

When the sample size is large, approximately 95% of the sample means taken from a population and same sample size will fall within ± 1.96 standard errors of the population mean, that is,

$$\mu \pm 1.96\left(\frac{\sigma}{\sqrt{n}}\right).$$

Now, if a specific sample mean is selected, say, , there is a 95% probability that the interval $\mu \pm 1.96\left(\frac{\sigma}{\sqrt{n}}\right)$ contains . Likewise, there is a 95% probability that the interval specified by

$$\overline{X} \pm 1.96\left(\frac{\sigma}{\sqrt{n}}\right)$$

will contain μ. Stated another way

$$\overline{X} - 1.96\left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \overline{X} + 1.96\left(\frac{\sigma}{\sqrt{n}}\right)$$

For a 99% confidence interval, the value 2.58 is used instead of 1.96 in the formula.

$$\overline{X} - 2.58\left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \overline{X} + 2.58\left(\frac{\sigma}{\sqrt{n}}\right)$$

Since other confidence intervals are used in statistics, the symbol $Z_{\alpha/2}$ is used in the general formula for confidence intervals.

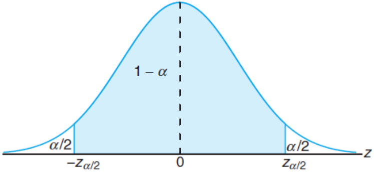
$$\overline{X} - Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \overline{X} + Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

According to the Central Limit Theorem, we can expect the sampling distribution of $\overline{X}$ to be approximately normally distributed with mean $\mu_{\overline{X}} = \mu$ and standard deviation $\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}$. Writing $z_{\frac{\alpha}{2}}$ for the z-value above which we find an area of $\frac{\alpha}{2}$ under the normal curve

$$P\left(-z_{\frac{\alpha}{2}} \leq Z \leq z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

$$\boxed{Z = \frac{\overline{X} - \mu}{\frac{\sigma}{\sqrt{n}}}}$$

$$P\left(-z_{\frac{\alpha}{2}} \leq \frac{\overline{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

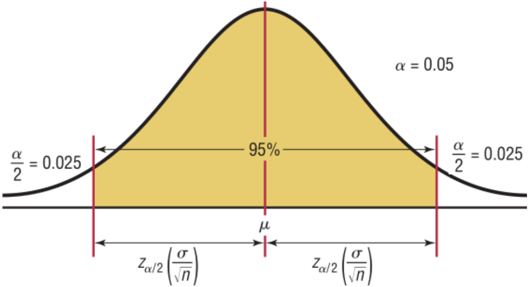


$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha.$

$$P\left(\overline{X} - z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} \quad \leq \quad \mu \quad \leq \quad \overline{X} - z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$\overline{X} - Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \overline{X} + Z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

α represents the total area in both tails of the standard normal distribution curve, and $\alpha/2$ represents the area in each one of the tails.



When the 95% confidence interval is to be found, $\alpha = 0.05$, since 1-0.05=0.95, or 95%.When $\alpha = 0.01$, then $1 - \alpha = 1 - 0.01 = 0.99$, and the 99% confidence interval is being calculated.

The average zinc concentration recovered from a sample of measurements taken in 36 different locations in a river is found to be 2.6 grams per milliliter. Find the 95% and 99% confidence intervals for the mean zinc concentration in the river. Assume that the population standard deviation is 0.3 gram per milliliter.

The point estimate of μ is $\bar{x} = 2.6$, $n = 36$ and $\sigma = 0.3$. Using this we have

$$2.6 - z_{\frac{\alpha}{2}} \frac{0.3}{\sqrt{36}} < \mu < 2.6 + z_{\frac{\alpha}{2}} \frac{0.3}{\sqrt{36}}$$

We can found α using

$$100\,(1-\alpha)\,\% = 95\% \qquad \Rightarrow (1-\alpha) = 0.95$$

This gives that $\alpha = 0.05$ and $\frac{\alpha}{2} = 0.025$. Now

$$z_{0.025} = 1.96 \qquad \text{using Normal distribution tabel.}$$

Thus the 95% confidence interval is

$$2.6 - (1.96)\left(\frac{0.3}{\sqrt{36}}\right) < \mu < 2.6 + (1.96)\left(\frac{0.3}{\sqrt{36}}\right) \implies 2.50 < \mu < 2.70$$

For 99% confidence interval $\alpha = 1 - 0.99 = 0.01$ and $\frac{\alpha}{2} = 0.005$. Hence

$$z_{0.005} = 2.575 \quad \text{using Normal distribution tabel.}$$

99% confidence interval is

$$2.6 - (2.575)\left(\frac{0.3}{\sqrt{36}}\right) < \mu < 2.6 + (2.575)\left(\frac{0.3}{\sqrt{36}}\right), \implies 2.47 < \mu < 2.73.$$

Example

A survey of 30 adults found that the mean age of a person's primary vehicle is 5.6 years. Assuming the standard deviation of the population is 0.8 year, find the best point estimate of the population mean and the 99% confidence interval of the population mean.[a]

[a] $\alpha = 1 - 0.99$ and $z_{\frac{\alpha}{2}} = 2.575$ with $n = 30$, $\mu = 5.6$ and $\sigma = 0.8$.

$$5.6 - 2.58\left(\frac{0.8}{\sqrt{30}}\right) < \mu < 5.6 + 2.58\left(\frac{0.8}{\sqrt{30}}\right)$$

$$5.6 - 0.38 < \mu < 5.6 + 0.38$$

$$5.22 < \mu < 5.98$$

$$5.2 < \mu < 6.0 \text{ (rounded)}$$

Hence, one can be 99% confident that the mean age of all primary vehicles is between 5.2 and 6.0 years, based on 30 vehicles.

When μ is known and the sample size is 30 or more, or the population is normally distributed if sample size is less than 30, the confidence interval for the mean can be found by using the z-distribution as

$$\overline{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} < \mu < \overline{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

If we have a random sample from a normal distribution, then the random variable

$$T = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}}$$

has a Student t-distribution. Here S is the sample standard deviation. In this situation, with σ unknown, T can be used to construct a confidence interval on μ.
The procedure is the same as that with μ known except that σ is replaced by S and the **standard normal distribution is replaced by the t-distribution**.

Formula for a Specific Confidence Interval for the Mean When σ Is Unknown and $n < 30$.

$$\overline{x} - t_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} < \mu < \overline{x} + t_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \text{ with degree of freedom } = n - 1.$$

Find the $t_{\frac{\alpha}{2}}$ value for a 95% confidence interval when the sample size is 22.

The t Distribution

	Confidence Intervals	50%	80%	90%	95%	98%	99%
d.f.	One tail α	0.25	0.10	0.05	0.025	0.01	0.005
	Two tails α	0.50	0.20	0.10	0.05	0.02	0.01
1							
2							
3							
⋮							
21					2.080	2.518	2.831
⋮							
(z)∞		0.674	1.282[a]	1.645[b]	1.960	2.326[c]	2.576[d]